

Equilibrium Distribution for t-Distributed Stochastic Neighbor Embedding with Generalized Kernel

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1 Introduction

The t-distributed stochastic neighbor embedding(t-SNE) is a well-known method of visualizing high-dimensional data with low-dimensional vectors (so they can be easily visualized). Let $d, s \geq 2$ with $d > s$, and $n \geq 1$. The input of t-SNE consist of vectors $X_1, \dots, X_n \in \mathbb{R}^d$. The goal is the find vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$ that serve as a good representation of the original vectors. The process of t-SNE starts with defining the following probability masses:

$$p_{j|i} = \frac{\exp(-\|X_i - X_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|X_i - X_k\|^2 / 2\sigma_i^2)}, \quad (1.1)$$

where σ_i are given constants. For each pair of data points X_i and X_j , we assign a weight to that pair, namely the Gaussian function $\exp(-\|X_i - X_j\|^2 / 2\sigma_i^2)$. We call such function the *input kernel*. In this paper, we are looking at generalized kernels. Let $\mathbf{Ker_I}(x, y, \sigma) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^+$ be such that $\mathbf{Ker_I}(x, y, \sigma) = \mathbf{Ker_I}(y, x, \sigma)$. We replace the probability mass in (1.1) with the following

$$p_{j|i} = \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)}. \quad (1.2)$$

σ_i satisfies that for each $i = 1, \dots, n$, that

$$-\sum_{j=1, j \neq i}^n p_{j|i} \log p_{j|i} = \log \mathbf{Prep} =: \log(n\rho). \quad (1.3)$$

$\mathbf{Prep}$ is called the perplexity of the model and $\rho \in (0, 1)$ is a given constant. The purpose of σ_i is to make sure that for each i , the probability distribution induced by $p_{j|i}$ has the same entropy across all i . Clearly, $p_{j|i}$ is not symmetric. We then define $p_{ii} = 0$ and for $1 \leq i \neq j \leq n$,

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} = \frac{1}{2n} \left(\frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)} + \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_j)}{\sum_{k \neq j} \mathbf{Ker_I}(X_j, X_k, \sigma_j)} \right). \quad (1.4)$$

Obviously,

$$\sum_{\substack{j=1 \\ j \neq i}}^n p_{j|i} = 1, \quad \sum_{1 \leq i \neq j \leq n} p_{ij} = 1. \quad (1.5)$$

As for the output vectors $Y_1, \dots, Y_n \in \mathbb{R}^s$, set for $1 \leq i \neq j \leq n$,

$$q_{ij} = \frac{\mathbf{Ker}_O(Y_i, Y_j)}{\sum_{k \neq l} \mathbf{Ker}_O(Y_k, Y_l)}. \quad (1.6)$$

The weight we assign to each pair of output data points Y_i and Y_j , a.k.a. a function $\mathbf{Ker}_O(Y_i, Y_j) : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}^+$, is called the output kernel. In the original algorithm, $\mathbf{Ker}_O(Y_i, Y_j) = (1 + \|Y_i - Y_j\|^2)^{-1}$. Denote $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ the input and output data set. The goal of t-SNE is to find Y that minimizes the relative entropy

$$L_{n,\rho}(X, Y) = \sum_{1 \leq i \neq j \leq n} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right). \quad (1.7)$$

More precisely, the output of t-SNE Y^* is defined as

$$Y^* = \arg \min_{Y \in (\mathbb{R}^s)^n} L_n(X, Y). \quad (1.8)$$

In the expression of $L_n(X, Y)$, all p_{ij} are given and q_{ij} is a function of Y_i and Y_j . Therefore, Y^* can be found simply by using gradient descent. This is no different from the original version of t-SNE. However, it is the mathematical properties and implications of t-SNE that we are interested in. Towards this end, let $\mathcal{M}(\mathbb{R}^d)$ be the space of Borel probability measures on $\mathbb{R}^d$ and $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ be a probability measure. Suppose all input $X = (X_1, \dots, X_n)$ are i.i.d., independent random variables with probability measure μ_X . Meanwhile, let $Y^* = (Y_1, \dots, Y_n)$ be the output of the t-SNE algorithm. Consider the following empirical measure:

$$\mu_n := \frac{1}{n} \sum_{i=1}^n \delta(X_i, Y_i) \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s). \quad (1.9)$$

We are interested in its limiting measure as $n \rightarrow \infty$. To finish the setup, there are a few more remaining quantity we would like to introduce. Let $\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s)$. For a given kernel $\mathbf{Ker}_I(x, y)$, define $F_{\rho,\mu}(x, \sigma) : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}$ as

$$F_{\rho,\mu}(x, \sigma) = \int \frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker}_I(x, x', \sigma)}{\int \mathbf{Ker}_I(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho. \quad (1.10)$$

Note that the last s variables in the integrands stay constant. It is not hard to see that $F_{\rho,\mu}(x, \sigma)$ is the continuous version of the entropy. Next, we extend the definition of σ_i by defining $\sigma_{\rho,\mu}^* : \mathbb{R}^d \rightarrow \mathbb{R}$ the unique solution of the equation

$$F_{\rho,\mu}(x, \sigma) = 0. \quad (1.11)$$

The introduction of σ^* aligns with the procedure of finding σ_i that controls the entropy to be the same for all i . Of course, the existence and uniqueness of σ^* is not trivial and will be discussed in section 4. For the symmetric probability mass, let $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ be a given function and $p_\psi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be defined by

$$p_\psi(x, x') = \frac{1}{2} \left(\frac{\mathbf{Ker}_I(x, x', \psi(x))}{\int \mathbf{Ker}_I(x, x', \psi(x)) d\mu(x')} + \frac{\mathbf{Ker}_I(x, x', \psi(x'))}{\int \mathbf{Ker}_I(x, x', \psi(x')) d\mu(x)} \right). \quad (1.12)$$

For the extension on the output side, let $q : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}$ be given by

$$q(y, y') = \frac{\mathbf{Ker}_O(y, y')}{\iint \mathbf{Ker}_O(y, y') d\mu(y) d\mu(y')}. \quad (1.13)$$

Based on the construction of p and q , define

$$I_\rho(\mu) = \iint p_{\sigma_{\rho,\mu}^*}(x, x') \log \left(\frac{p_{\sigma_{\rho,\mu}^*}(x, x')}{q(y, y')} \right) d\mu(x, y) d\mu(x', y'). \quad (1.14)$$

Recall $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ is an arbitrary given probability measure. We set $\mathcal{P}_X$ to be the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^s$ whose marginal coincides with μ_X . Formally speaking,

$$\mathcal{P}_X := \{\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s) : \text{the } \mathbb{R}^d \text{ marginal of } \mu \text{ is } \mu_X\}. \quad (1.15)$$

Lastly, define

$$\tilde{\mathcal{P}}_X := \left\{ \mu \in \mathcal{P}_X : \int (p_{\sigma_{\rho,\mu}^*}(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} \mathbf{Ker}_O(y, y')}{\mathbf{Ker}_O(y, y')} d\mu(x', y') = 0, \mu(x, y)\text{-a.e.} \right\}. \quad (1.16)$$

2 Main Results

The core results of this problem are the conditions we assume the input and output kernels to have such that the output of the generalized t -SNE algorithm converges to a limiting equilibrium measure. Towards this end, below are the constraints we impose on the input and output kernels.

Condition 1 (Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of*

$$\mathbf{Ker}_I(x, x', \sigma) = \exp(-w(\sigma \|x - x'\|^\theta)) \quad (2.1)$$

Here, σ is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. Then w satisfies:

1. $\forall t \geq 0, w'(t) > 0, \lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. The following integral converges:

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty; \quad (2.2)$$

Since the all probability masses are normalized, we can assume without loss of generality that $w(0) = 0$ from now on.

Condition 2 (Output Kernel Conditions). *The output kernel depends only on the distance between two output data points y and y' . In particular, $\mathbf{Ker}_O(y, y') = k(\|y - y'\|)$ and that:*

1. $k : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^+$ is decreasing and bounded;
2. $\mathbf{Ker}_O$ satisfies

$$\iint_{(0,\infty)^2} \mathbf{Ker}_O(y, y') dy dy' < \infty; \quad (2.3)$$

3. k has bounded derivative;
4. $k'(0) = 0$.

We are ready to state the main results

Theorem 1. Assume μ_X has a C^1 density $f(x)$ and compact support. Let $X_i \in \mathbb{R}^d$ be i.i.d random variables with probability measure μ_X . Then for any $\rho \in (0, 1)$,

$$\lim_{n \rightarrow \infty} \inf_Y L_{n,\rho}(X, Y) = \inf_{\mu \in \bar{\mathcal{P}}_X} I_\rho(\mu). \quad (2.4)$$

Moreover, there exist a measure μ^* and a sub-sequence $\{n_k\}$ such that $\mu_{n_k} \rightarrow \mu^*$ weakly and $I_\rho(\mu^*) = \inf_{\mu \in \bar{\mathcal{P}}_X} I_\rho(\mu)$.

Theorem 2. Under the same assumptions of theorem 1, μ^* has compact support.

Remark. From section 5 up to a very late point in our proofs, we will prove our lemmas under lighter assumptions on the probability measure μ_X we draw the input samples from. We will assume μ_X is sub-Gaussian rather than with compact support. For results in those parts to be true, we need to add one more constraint to the input kernel, as shown in the **condition below**. The reader can easily verify that if we assume μ_X to be compact from the beginning, all the lemmas can be easily proven using the exact same framework (and in fact, easier). We will discuss why we choose to prove those lemmas under the sub-Gaussian assumption, as well as other intuition and insights in section 3.

Condition 3 (Input Kernel Conditions). Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\mathbf{Ker}_I(x, x', \sigma) = \exp(-w(\sigma \|x - x'\|^\theta))$. Here, σ is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. Define

$$\mathcal{K}_w = \left\{ \tilde{C} \in \mathbb{R}_{\geq 0} \left| \lim_{n \rightarrow \infty} n^{-1/2} (\log n)^\theta \exp \left(4w \left(\left(\tilde{C} \sqrt{\log n} \right)^\theta \right) \right) = 0 \right. \right\}. \quad (2.5)$$

Then w satisfies:

1. $\forall t \geq 0, w'(t) > 0, \lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. The following integral converges:

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty; \quad (2.6)$$

4. $\exists M > 0$ s.t. $\forall t \geq 0, w'(t) \leq M$, and $\sup \mathcal{K}_w > 0$.

3 Intuitive Insights and Simulations

We dedicate this section to provide some intuition and insights on the context. From the form of the kernels we choose, to why we introduce the extra condition 3, and to a series of simulation plots that demonstrate the effectiveness of the generalized t -SNE algorithm. we hope to provide the reader with a better understanding of the problem and the results we aim to prove.

Is the form of the input kernel general enough?

The input kernel may be the most important part of the algorithm. Since we would like to expand the result of the original version, it is important that we do not lose comprehensiveness when assuming a concrete form of the input kernel. Therefore, (2.1) is due an explanation.

We start by realizing that t -SNE is about the relationship between each data point. The insights we gain from the algorithm should be translation and rotation invariant with respect to the coordinate systems we choose. Hence, it is only reasonable to assume that the input kernel is a function of $\|x - x'\|$ for each pair of input data points x, x' .

On the other hand, the weight we assign to each pair of data points is always positive due to nature of probability mass. We thus lose no generality by assuming the input kernel has an exponential form because obviously, the exponential function is a smooth bijection from $\mathbb{R}$ to $\mathbb{R}^+$.

Additionally, though it is coincidental that in the original version of the algorithm σ_i can be viewed as the variance of a Gaussian random variable, in reality σ_i 's are only used to control the entropy of each probability distribution induced by x_i . With that being said, we only need to introduce σ in such a way that it is able to control the entropy for each x . As we can see in section 4, simply multiplying σ to $\|x - x'\|^\theta$ is enough to achieve this goal. Moreover, we prove the convergence of σ^* in section 5, providing the anchor of the following results just as in [Auffinger and Fletcher \[2023\]](#).

Finally, combining all the above, we arrive at the form of the input kernel in (2.1), by adding a general function w to the product $\sigma \|x - x'\|^\theta$. This gives us a kernel we can easily work on without losing any generality.

Are the conditions on the input/output kernel general enough?

Besides the form of the input kernel, we also want to make sure that the conditions we impose on the kernels are also general enough for the significance of our results. Since we frequently deal with integrals containing the input and output kernels, their integrability is the bare minimum we should assume. Along with integrability comes their decay to 0 at infinity. Apart from that, we do not add any excessive constraints to the kernels and readers will see later in proofs that those assumptions are strictly necessary for the results to hold.

Specifically, notice the second assumption of condition 1 is basically requiring the input kernel to decay faster than a minimal speed by controlling its second order derivative. If we let $w(t) = \log t$, we have $w'(t) + tw''(t) = 0$. Roughly speaking, this implies that the main theorems hold as long as the input kernel decays at polynomial rate or faster, which is a big step up than proving the convergence for Gaussian kernels alone.

Why do we introduce condition 3?

[Yi: Ask Tuca about this and come back.]

Simulations

Another important question we would like to ask, is that can the results in the main theorems be visualized and thus verified? In theory, if we choose any input and output kernels meeting condition 1 and 2 respectively, we are able to get an asymptotic equilibrium measure and each equilibrium measure will look different for different combinations of input and output kernels.

Towards that end, we choose four different combinations of input and output kernels, including the original one, and run t -SNE algorithms on the exactly same MNIST data set. As shown in figure 1, we used four different configurations for the input and output kernels. They are

(a) Original t -SNE kernels;

(b) Polynomial input kernel $\mathbf{Ker_I}(x_i, x_j, \sigma_i) = (\sigma_i \|x_i - x_j\|^{10} + 1)^{-1}$;

(c) Mixed input kernel $\mathbf{Ker_I}(x_i, x_j, \sigma_i) = (\sigma_i \|x_i - x_j\| + 1)^{-10} + \exp(-\sigma_i \|x_i - x_j\|)$;

(d) Custom input and output kernels with

$$\begin{aligned}\mathbf{Ker_I}(x_i, x_j, \sigma_i) &= (\sigma_i \|x_i - x_j\| + 1)^{-10} + \exp(-\sigma_i \|x_i - x_j\|), \\ \mathbf{Ker_O}(y_i, y_j) &= (1 + \|y_i - y_j\|^2 + \|y_i - y_j\|^4)^{-1}.\end{aligned}$$

In the end, all four combinations are able to converge and produce on-par performance, in terms of distinguishing different digits. Meanwhile, each combination shows a different distribution of the output data points, which is consistent with our theoretical results.

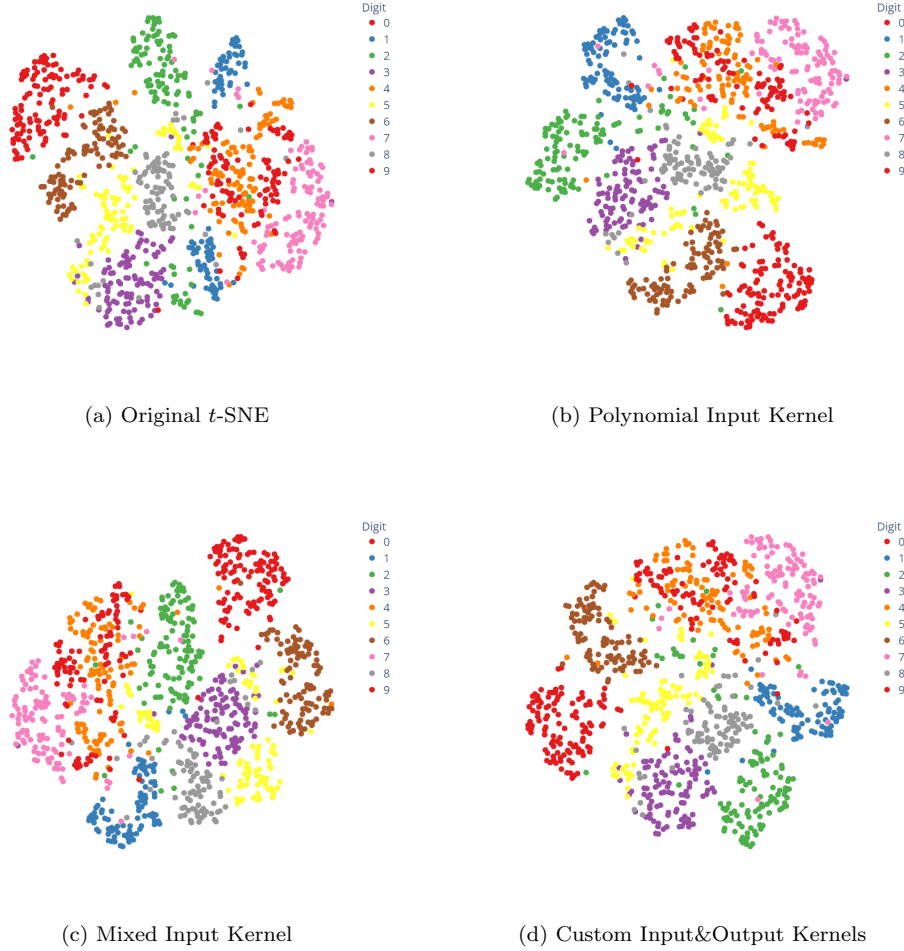


Figure 1: t -SNE running with different combinations of input and output kernels on the MNIST data set. All three custom kernel combinations are able to perform on par with the original algorithm.

4 Existence and Uniqueness of σ^*

In this section we establish the existence and uniqueness of $\sigma_{\rho,\mu}^*(x)$. Recall that $\sigma_{\rho,\mu}^*(x)$ is the unique solution of the equation

$$F_{\rho,\mu}(x, \sigma) = \int \frac{\mathbf{Ker}_{\mathbf{I}}(x, x', \sigma)}{\int \mathbf{Ker}_{\mathbf{I}}(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker}_{\mathbf{I}}(x, x', \sigma)}{\int \mathbf{Ker}_{\mathbf{I}}(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho = 0 \quad (4.1)$$

Intuitively speaking, σ^* is the limit function of σ_i 's that are used to control the entropy. The purpose of this section is to prove that such limit exists and is unique for each given x . The existence and uniqueness of σ^* will become the foundation of everything we will prove afterwards. By condition 1, we can write

$$\begin{aligned} F_{\rho,\mu}(x, \sigma) &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (4.2)$$

We start with the lemma that gives F an increasing lower bound.

Lemma 1. *If μ has a C^1 density $f(x)$ and both f and $\|\nabla f\|$ are bounded, then there exists a constant $C > 0$ such that*

$$F_{\rho,\mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \quad (4.3)$$

Proof. Notice that for any integrable function h on $\mathbb{R}_{\geq 0}$, we have

$$\int_{\mathbb{R}^d} h(\|x\|) dx = \int_{\mathbb{S}^{d-1}} \int_0^\infty r^{d-1} h(r) dr d\nu = S_{d-1} \int_0^\infty r^{d-1} h(r) dr, \quad (4.4)$$

where $\mathbb{S}^{d-1}$ is the $(d-1)$ -dimensional sphere, $d\nu$ the corresponding surface measure, and S_{d-1} the volume of the sphere. By Condition 1, we know that

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (4.5)$$

Since w is increasing and $\lim_{t \rightarrow \infty} w(t) = \infty$, it is easy to see that

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty. \quad (4.6)$$

Denote

$$Z_d := \int_{\mathbb{R}^d} \exp(-w(\|\tau\|^\theta)) d\tau = S_{d-1} \int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt. \quad (4.7)$$

Consider

$$\begin{aligned} &\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') = \int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx' \\ &= Z_d \sigma^{-d/\theta} \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx'. \end{aligned} \quad (4.8)$$

Because $\lim_{x \rightarrow \infty} w(x) = \infty$, the integrand of the above integral converges weakly to $\delta_x f(x')$. To prove this statement, notice that

$$\begin{aligned}
& \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{Z_d \sigma^{-d/\theta}} f(x') dx' \\
&= \frac{1}{Z_d} \int \exp\left(-w\left(\|\sigma^{1/\theta}(x - x')\|^\theta\right)\right) f(x') d\sigma^{1/\theta}(x - x') \\
&\equiv \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau.
\end{aligned} \tag{4.9}$$

Since f is bounded, $\exp(-w(\|\tau\|^\theta)) \sup f$ is integrable. By dominant convergence theorem and the continuity of f , we have

$$\begin{aligned}
& \lim_{\sigma \rightarrow \infty} \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\
&= \frac{1}{Z_d} \int \lim_{\sigma \rightarrow \infty} \exp(-w(\|\tau\|^\theta)) f(x - \tau \sigma^{-1/\theta}) d\tau \\
&= f(x) \frac{1}{Z_d} \int \exp(-w(\|\tau\|^\theta)) d\tau \\
&= f(x).
\end{aligned} \tag{4.10}$$

There exists some $\epsilon(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$ and a constant $C > 0$, that

$$\begin{aligned}
-\log\left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')\right) &= -\log\left(Z_d \sigma^{-d/\theta}\right) - \log f(x) + \epsilon(\sigma) \\
&\geq C \log \sigma - \log f(x).
\end{aligned} \tag{4.11}$$

Next, since f is bounded, there exists $M > 0$ such that $f \leq M$.

$$\begin{aligned}
& \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\
&\leq M \frac{\int w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta)) dx'}{\int \exp(-w(\sigma \|x - x'\|^\theta)) f(x') dx'} \\
&= M \frac{\sigma^{-d/\theta} \int w(\|\sigma^{1/\theta}(x - x')\|^\theta) \exp(-w(\|\sigma^{1/\theta}(x - x')\|^\theta)) d\sigma^{1/\theta}(x - x')}{\int \exp(-w(\sigma \|(x - x')\|^\theta)) f(x') dx'} \\
&= M \frac{\sigma^{-d/\theta} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau}{\int \exp(-w(\sigma \|(x - x')\|^\theta)) f(x') dx'} \\
&\rightarrow \frac{M}{Z_d f(x)} \int w(\|\tau\|^\theta) \exp(-w(\|\tau\|^\theta)) d\tau < \infty.
\end{aligned} \tag{4.12}$$

The numerator converges because of condition 1. As the bound in (4.12) does not depend on σ , we can combine (4.2), (4.11) and (4.12) and deduce that

$$F_{\rho, \mu}(x, \sigma) \geq C \log \sigma - \log f(x) + \log \rho. \tag{4.13}$$

□

This lemma proves the existence of $\sigma_{\rho, \mu}^*(x)$. Indeed, by (4.2), apparently

$$\lim_{\sigma \rightarrow 0} F_{\rho, \mu}(x, \sigma) = \log \rho < 0 \tag{4.14}$$

since $\rho \in (0, 1)$. Now with lemma 1, we get

$$\lim_{\sigma \rightarrow \infty} F_{\rho, \mu}(x, \sigma) = \infty. \quad (4.15)$$

By continuity of f and intermediate value theorem, $\sigma_{\rho, \mu}^*(x)$ exists. To prove the uniqueness, we are left to show that $F_{\rho, \mu}$ is strictly increasing. The next lemma will do just that.

Lemma 2. *Given the same condition as in lemma 1, $F_{\rho, \mu}(x, \sigma)$ is strictly increasing in σ .*

To tackle this proof, we need one additional lemma inspired by theorem 7 in [Liu et al., 2009].

Lemma 3. *Let $f(x), g(x), h(x)$ be continuous functions $\mathbb{R}^d \rightarrow \mathbb{R}^+$ such that for all $x, y \in \mathbb{R}^d$*

$$(g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) \geq 0. \quad (4.16)$$

Let μ be an arbitrary Lebesgue measure and assume all integrals in this lemma converge. Then the inequality

$$\frac{\int_{\mathbb{R}^d} f(x) d\mu}{\int_{\mathbb{R}^d} h(x) d\mu} \geq \frac{\int_{\mathbb{R}^d} f(x) g(x) d\mu}{\int_{\mathbb{R}^d} h(x) g(x) d\mu} \quad (4.17)$$

holds. If (4.16) reverses, then (4.17) reverses.

Proof. The proof of this lemma is similar to the one in [Liu et al., 2009]. It suffices to show that

$$\int_{\mathbb{R}^d} f(x) d\mu \int_{\mathbb{R}^d} h(x) g(x) d\mu \geq \int_{\mathbb{R}^d} h(x) d\mu \int_{\mathbb{R}^d} f(x) g(x) d\mu. \quad (4.18)$$

The above inequality is equivalent to

$$\int_{\mathbb{R}^d} f(x) d\mu(x) \int_{\mathbb{R}^d} h(y) g(y) d\mu(y) \geq \int_{\mathbb{R}^d} h(x) d\mu(x) \int_{\mathbb{R}^d} f(y) g(y) d\mu(y). \quad (4.19)$$

That means we need to show

$$\mathcal{G} := \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(y) (f(x) h(y) - f(y) h(x)) d\mu(x) d\mu(y) \geq 0. \quad (4.20)$$

Swap x and y in the formula of $\mathcal{G}$ and we have an equivalent expression:

$$\mathcal{G} = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(x) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y). \quad (4.21)$$

Adding the last two formulas together yields

$$\begin{aligned} 2\mathcal{G} &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (g(x) - g(y)) (f(y) h(x) - f(x) h(y)) d\mu(x) d\mu(y) \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} h(x) h(y) (g(x) - g(y)) \left(\frac{f(y)}{h(y)} - \frac{f(x)}{h(x)} \right) d\mu(x) d\mu(y) \geq 0 \end{aligned} \quad (4.22)$$

by the hypothesis of the lemma. This finishes the proof. $\square$

Proof of Lemma 2. We compute the derivative of $F_{\rho, \mu}(x, \sigma)$ directly using (4.2). The following simplified notations will be used in the first several steps:

$$w = w(\sigma \|x - x'\|^\theta), \quad w' = w'(\sigma \|x - x'\|^\theta), \quad \exp(\cdot) = \exp(-w(\sigma \|x - x'\|^\theta)). \quad (4.23)$$

Omit $d\mu(x')$ in the integration for cleaner writing, and we have

$$\begin{aligned}
& \frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \\
&= - \int \frac{w' \|x - x'\|^\theta \exp(\cdot)}{\int \exp(\cdot)} - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} - \frac{(\int \exp(\cdot))'}{\int \exp(\cdot)} \\
&= - \int \frac{w \exp(\cdot)'}{\int \exp(\cdot)} + \int \frac{w \exp(\cdot) (\int \exp(\cdot))'}{(\int \exp(\cdot))^2} \quad (\text{The first and fourth terms cancel.}) \\
&= \int \frac{w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot)}{\int \exp(\cdot)} - \frac{\int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot)}{(\int \exp(\cdot))^2}. \tag{4.24}
\end{aligned}$$

We are able to move differentiation inside the integral because of dominant convergence theorem. To show that $\frac{\partial}{\partial \sigma} F_{\rho, \mu}(x, \sigma) \geq 0$, we just need to show

$$\int w \cdot w' \cdot \|x - x'\|^\theta \cdot \exp(\cdot) \int \exp(\cdot) \geq \int w \cdot \exp(\cdot) \int w' \cdot \|x - x'\|^\theta \exp(\cdot). \tag{4.25}$$

For each $x, y, x' \in \mathbb{R}$ if the equality

$$\begin{aligned}
& (w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta) \\
& \cdot \left(\frac{1}{w(\sigma \|y - x'\|^\theta)} - \frac{1}{w(\sigma \|x - x'\|^\theta)} \right) \geq 0 \tag{4.26}
\end{aligned}$$

holds, then by letting $f(x) = \exp(\cdot)$, $h(x) = w \cdot \exp(\cdot)$ and $g(x) = w' \cdot \|x - x'\|^\theta$, we have (4.25) to be true by lemma 3. Let $\|x - x'\|^\theta > \|y - x'\|^\theta$ without loss of generality. Then by monotonicity of w , we have $w(\sigma \|x - x'\|^\theta) > w(\sigma \|y - x'\|^\theta)$. In order for (4.26) to be true, we are left to show

$$w'(\sigma \|x - x'\|^\theta) \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \|y - x'\|^\theta \geq 0. \tag{4.27}$$

Since $\sigma > 0$, the above inequality is equivalent to

$$w'(\sigma \|x - x'\|^\theta) \sigma \|x - x'\|^\theta - w'(\sigma \|y - x'\|^\theta) \sigma \|y - x'\|^\theta \geq 0. \tag{4.28}$$

Let $\tilde{w}(x) = xw'(x)$. Based on the second assumption in condition 1 and mean value theorem, $\tilde{w}$ satisfies for all $a, b \geq 0$, $a > b$, there exists $c \in [b, a]$ that

$$\frac{aw'(a) - bw'(b)}{a - b} = w'(c) + cw''(c) \geq 0. \tag{4.29}$$

This proves (4.28) by letting $a = \sigma \|x - x'\|^\theta$ and $b = \sigma \|y - x'\|^\theta$, as desired. $\square$

We can formally state and prove the main result of this section

Proposition 1. *Let $\mu \in \mathcal{M}(\mathbb{R}^d)$ with C^1 density f such that both f and $\|\nabla f\|$ are bounded. The equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. The existence and uniqueness can be proven by combining (4.14), (4.15), lemma 1 and lemma 2. The smoothness of $\sigma_{\rho, \mu}^*(x)$ follows from implicit function theorem. $\square$

5 Uniform Convergence of σ^*

6 Proofs of the Main Results

Lemma 4. *As $n \rightarrow \infty$, we have*

$$\iint g(y, y') d\mu_n(y) d\mu_n(y') - \frac{1}{n} = \iint g(y, y') d\mu^* d\mu^* + o(1). \quad (6.1)$$

7 Extra Technical Proofs

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